

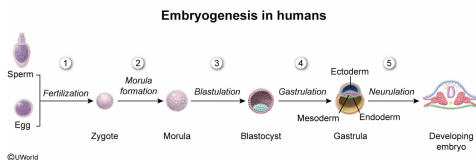
Which series shows the order in which a human embryo develops?

- ☐ A. Fertilization → blastulation → morula formation → gastrulation → neurulation [13%]
- ☐ B. Fertilization → morula formation → blastulation → neurulation → gastrulation [11%]
- ✓ ☒ C. Fertilization → morula formation → blastulation → gastrulation → neurulation [71%]
- ☐ D. Fertilization → neurulation → morula formation → blastulation → gastrulation [2%]

Correct

70% answered correctly

Explanation:



In humans, embryogenesis occurs in the following order:

1. **Fertilization:** The fusion of a sperm with an oocyte results in a single-celled zygote.
2. **Morula formation:** The zygote undergoes successive mitotic divisions (cleavage) to form a ball of cells known as a morula.
3. **Blastulation:** The morula is transformed into a hollow ball of cells known as the blastocyst. At this stage, the developing embryo implants in the uterine wall.
 - a. The blastocyst consists of two layers, the trophoblast (or hollow cell sphere) and the inner cell mass, which is clustered inside of the trophoblast.
4. **Gastrulation:** The double-layered blastocyst becomes a triple-layered gastrula. It is through gastrulation that a developing embryo forms germ layers (ie, ectoderm, mesoderm, endoderm). Gastrulation begins with the formation of a primitive structure (primitive streak) that facilitates cell migration on the bilaminar disk, a two-layer cell sheet that forms in the trophoblast. Each of the three germ layers then continues to develop into specific structures of the body.
5. **Neurulation:** At this stage, the nervous system forms. The notochord, a structure deriving from the mesoderm, releases signals causing the ectoderm above it to thicken and form the neural plate. The neural plate eventually folds inward, forming the neural groove, and the neural folds (edges of the neural plate) join to form the neural tube.
 - a. The neural tube develops into the central nervous system (brain and spinal cord).
 - b. The remaining portions of the neural folds are known as the neural crest. These cells migrate from the neural tube and eventually give rise to most of the peripheral nervous system.

Accordingly, the correct order in which a human embryo develops is **fertilization → morula formation → blastulation → gastrulation → neurulation**.

Educational objective:

Embryogenesis is the process by which an embryo develops from a single-celled zygote. It includes the following stages in order: fertilization, morula formation, blastulation, gastrulation, neurulation.

Time spent: 0 sec

Subject:
Biology

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System:
Reproduction